Problem List

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1658. Minimum Operations to Reduce X to Zero

Solved

Medium | Topics | Companies | Hint

You are given an integer array `nums` and an integer `x`. In one operation, you can either remove the leftmost or the rightmost element from the array `nums` and subtract its value from `x`. Note that this **modifies** the array for future operations.

Return *the **minimum number** of operations to reduce* `x` *to **exactly*** `0` *if it is possible, otherwise, return* `-1`.

Example 1:

```
Input: nums = [1,1,4,2,3], x = 5
Output: 2
Explanation: The optimal solution is to remove the last two elements to reduce x to zero.
```

Example 2:

```
Input: nums = [5,6,7,8,9], x = 4
Output: -1
```

Example 3:

```
Input: nums = [3,2,20,1,1,3], x = 10
```

5.8K | 96 | 18 Online

Code

C++ | Auto

```cpp
        int total=0;
        for(int n:nums){
            total+=n;
        }
        int target=total-x;
        if(target<0)return -1;
        if(target==0) return nums.size();
        int left=0;
        int sum=0;
        int maxlen=-1;
        for(int right=0;right<nums.size();right++){
            sum+=nums[right];
            while(sum>target){
                sum-=nums[left];
                left++;
            }
            if(sum==target){
                maxlen=max(maxlen,right-left+1);
            }
        }
        if(maxlen==-1) return -1;
        return nums.size()-maxlen;
    }
};
```

Saved

Ln 16, Col 18

Testcase | Test Result